



MAN Diesel

S50MC-C

**VOLUME II
MAINTENANCE**

Instructions for Main Engines

This book forms part of a set of books consisting of three volumes entitled:

- Vol. I OPERATION
- Vol. II MAINTENANCE
- Vol. III COMPONENTS, DESCRIPTIONS

The purpose of these books is to provide general guidance on operation and maintenance and to describe the constructional features of a standard version of an MAN B&W main diesel engine. Deviations may be found in a specific plant. In addition, the books can be used for reference purposes, for instance in correspondence and when ordering spare parts.

It is essential that the following data is stated in spare parts orders as it is used by us to ensure the supply of the correct parts for the individual engines:

1. Name of vessel
2. Engine No. built by
3. Plate No.
4. Part No.
5. Quantity required (and description)

Example: M/S Nybo – 7730 B&W – P90201-0036 – 059
 10 off (piston ring)

★)The Plate No. may consist of either of these two sets of digits, as the pages are in the process of being updated with new numbers: P90201-0036 or 90201-036.

To ensure optimum efficiency, reliability and lifetime of the engine and its components, only original spare parts should be used when replacing parts on the engine.

When used in texts and illustrations, the designation “D” refers to the information given on the data sheets inserted in the respective books.

Reliable and economical operation of the diesel engines is conditional upon its correct operation and maintenance in accordance with MAN B&W Diesel A/S' instructions. Emissions related maintenance of the diesel engine shall be performed as specified in MAN B&W Diesel A/S' instructions and any additional instructions to that effect included in the Technical File. Consequently, it is essential that the engine room personnel are fully acquainted with the contents of this book and, in respect of instructions on emissions related maintenance of the diesel engine, also the additional instructions to that effect set out in the Technical File.

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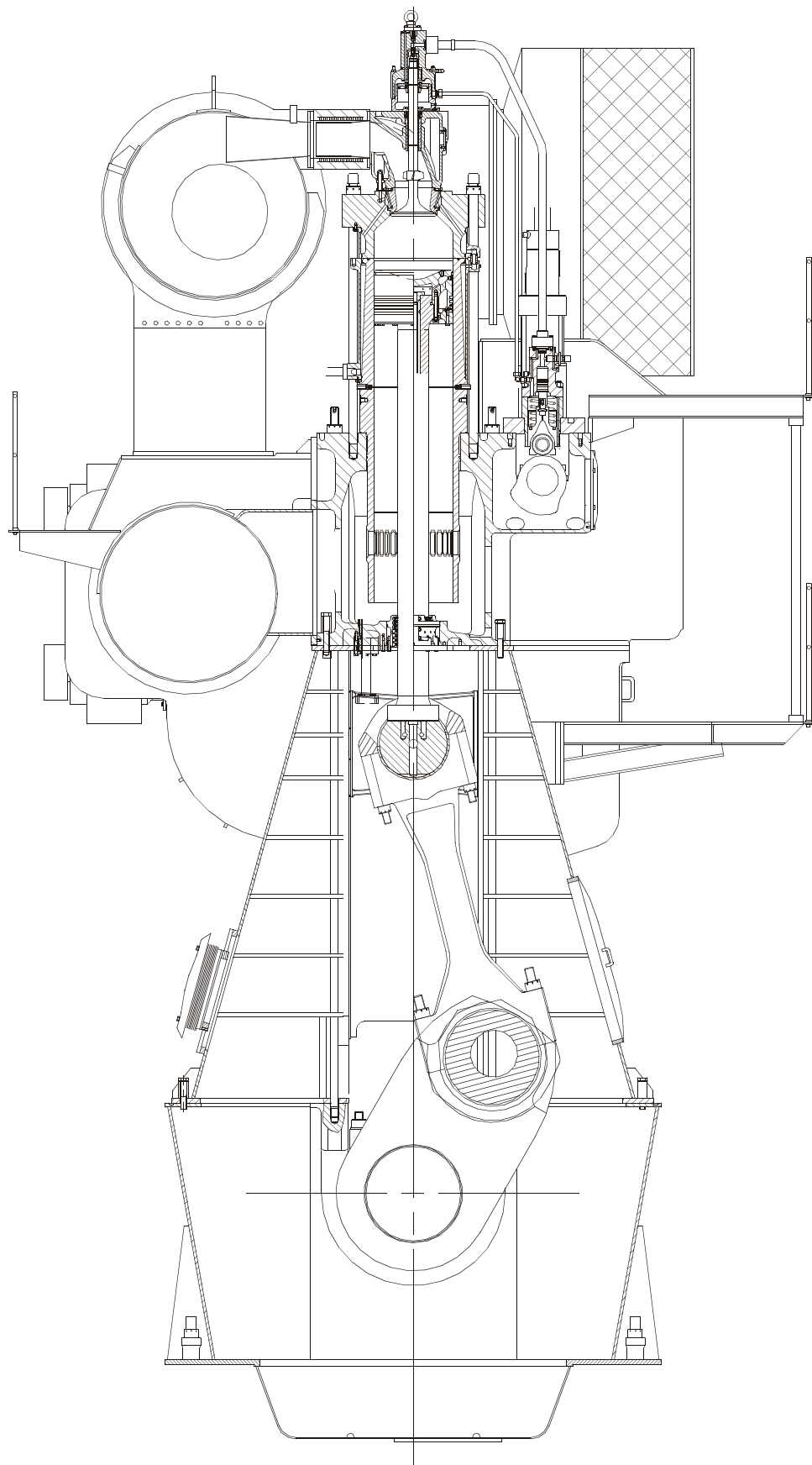
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The Checking and Maintenance Schedules indicate the intervals at which it is deemed appropriate to inspect the individual components of the engine and to carry out overhauls, if necessary, based on the engine condition or a criterion of time.

The stated **Regular checks** or **Service interval** are intended primarily as a guide, as differences in the actual service conditions, the quality of the fuel oil or lubricating oil, the treatment of cooling water, etc., will decisively influence the actual service results, and thus the intervals between necessary overhauling.

Design modifications may necessitate a revision of the instructions, in which case the revised instructions and changed overhauling intervals, if any, will apply and supersede those originally issued (see *e.g. our Service Letters*).

In addition to the checking and overhauling intervals stated in this schedule, please note that the periodical survey requirements of the classification society may require additional checks and reference is made to be carried out. For further information reference is made to the classification society.

The procedures are divided into five categories:

Time-based condition checking procedures

marked with a **C**, under the heading **Regular checks** or **Service interval**, deal with the service condition of a number of engine components, and form the basis for estimating whether further overhauling is necessary. In a number of cases the condition checking procedures refer to **Volume I** of the instruction book, in which more detailed descriptions and working procedures can be found.

Condition-based overhauling procedures

are marked with an **O**, under the heading **Regular checks** or **Service interval**. Under the heading **Refer to** (column P), a reference to additional information is stated.

This procedure number normally refers to one of the above condition checking procedures which form the basis of the overhaul. For this reason, the intervals stated are for guidance only.

Time-based overhauling procedures

also marked with an **O**, under the headings **Regular checks** or **Service interval** or **Based on observations**, are the procedures where an actual basis for estimation is lacking. It is recommended, therefore, to carry out these procedures at the overhauling intervals stated as a basis.

The letters **O** or **C** in **Based on observations** (column B), indicate that special service conditions may make checking or overhauling necessary beyond the actual standard schedules indicated.

(continued)

Initial checking procedures

marked with a **C**, under the heading **Check new/overhauled parts at 500, 1000 and 1500 hours** (column H), deal with initial checks that must be carried out on a new engine, or when parts have been replaced.

After the three initial checks have been completed, checks need only be carried out as per heading **Regular checks** or **Service interval**.

When using **electronic** maintenance planning systems, special attention should be paid to all initial checking procedures, as handling of such procedures often can not be done automatically. Instead such procedures must be handled manually.

Initial adjustment procedures

marked with an **A**, under the heading **Check new/overhauled parts at 500, 1000 and 1500 hours** (column H), deal with initial adjustment that must be carried out on a new engine, or when parts have been replaced.

After the three initial adjustments have been completed, adjustments need only be carried out as per headings **Regular checks** or **Service interval**.

When using **electronic** maintenance planning systems, special attention should be paid to all initial adjustment procedures, as handling of such procedures often can not be done automatically. Instead such procedures must be handled manually.

NOTE!

Special attention should be paid to the chain tightener, which **must** be adjusted three times (at 500, 1000 and 1500 hours of operation) on a new engine, or when chains or chain wheels have been replaced.

V : See Volume I "Operation"
 A : Adjustment to be carried out
 C : Check the condition

M : See maker's instructions
 O : Overhaul to be carried out
 H : Check new/overhauled parts at 500, 1000 and 1500 hours

R : Parts to be replaced
 B : Based on observations
 P : Refer to

No.	Procedure	H	Regular Checks			Service interval (x 1000 hours of operation)								B	P	Remarks
			Daily	Weekly	Monthly	2	4	6	8	12	16	24	32			
901	Cylinder Cover															
-1	Cylinder cover										C*			O		* Check whenever exhaust valve is removed
	Indicator cock													O		
902	Piston with Rod and Stuffing box															
-1	Piston									C,O				O		* Check clearances without dismantling segments
	Inspection through scavenge ports				C										V	
-2	Piston rod stuffing box									C*				O		
903	Cylinder Liner and Cylinder Lubrication															
-1	Cylinder liner													O		
	Inspection through scavenge ports				C											
	Measuring wear inside liner									C						
-2	Cylinder lubricators								C					A	V,M	
	Accumulators (Only for Alpha lubricator)								C				O	A		
904	Crosshead with connecting rod															
-1	Crosshead bearing	C							C					C,O	V	Bearings should only be opened if bearing material fragments fall out.
-2	Crosshead	C							C					C,O		
-3	Reciprocating parts								C					C,O		
-4	Crankpin bearing	C							C					C,O	V	
-5	Connecting rod													O		
905	Crankshaft and Thrust Bearing															Bearings should only be opened if bearing material fragments fall out.
-2	Main bearing	C							C					C,O	V	
-3	Thrust bearing	C							C					C,O		
-3	Guide bearing	C							C					C,O		
-4	Journal bearing	C							C					C,O	V	
-5	Axial vibration damper													O		
	Electronic axial vibration monitor (If installed)		C													
	Mechanical checking								C							

V : See Volume I "Operation"
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No.	Procedure	H	Regular Checks			Service interval (x 1000 hours of operation)								B	P	Remarks
			Daily	Weekly	Monthly	2	4	6	8	12	16	24	32			
-7	Tacho pick-up													C,O	M	New or overhauled chain to be checked/retightened after 500, 1500 and 4000 hours
-8	Angle encoder (Only for Alpha lubricator)													C,O	M	
906	Mechanical Control Gear															
-1	Chains	C					C							R	V	
-2	Chain tightener	C,A					C,A							A		
	Chain wheels, spray nozzles and guide bars	C					C							R		
-3	Camshaft	C							C					C,A		
-4	Camshaft bearings								C					C,O		
	Camshaft coupling, fitted bolts								C							
-5	Moment compensator	C					C							C,A		
	Governor						O								M	
	Functional check of overspeed device								C						M	
	Functional check of speed-setting system (engine with bridge control system)								C						M	
907	Starting Air System															
-1	Starting air distributor	C								C				A		
-2	Starting air valve									C,O						
908	Exhaust Valve															
-1	High-pressure pipe													O		
-2	Exhaust valve															
	Conventional hard-faced valve seats							O								
	Nimonic exhaust valves										O					
-3	Exhaust valve actuator												O			
	Actuator safety valve								C,A							
-4	Exhaust valve roller guide	C*				C'							C	O		* Only inspection through camshaft covers
-5	Exhaust valve cam	C*				C*								O		
-7	Exhaust valve special running													O		

V : See Volume I "Operation"			M : See maker's instructions											R : Parts to be replaced		
A : Adjustment to be carried out			O : Overhaul to be carried out											B : Based on observations		
C : Check the condition			H : Check new/overhauled parts at 500, 1000 and 1500 hours											P : Refer to		
No.	Procedure	H	Regular Checks			Service interval (x 1000 hours of operation)								B	P	Remarks
			Daily	Weekly	Monthly	2	4	6	8	12	16	24	32			
909	Fuel Oil System															
	Fuel oil water content			C												
-1	Fuel pump settings													A,C	V	
-2	VIT system (If applied)													A	V	
-3	Fuel pump cam													A		
-4	Fuel pump						C				O			O		
-5	Fuel pump top cover													O		
-6	Fuel pump barrel assembly													R		
-7	Fuel pump suction valve								C,O							
-8	Fuel pump puncture valve								C,O							
-9	Fuel pump shock absorber								C					O		
-9	Fuel system shock absorber								C					O		
-10	Lifting gear for roller guide								C					O		
-11	Fuel valve								C,O							
-	Fuel nozzle										R*			R		* HIP compound type nozzles only. Other types: 8000 hours
-12	Spindle guide										R			R		
-13	Non-return valve								O							
-14	Fuel oil high-pressure pipes													O		* Only inspection through camshaft covers
-15	Fuel pump roller guide	C*							C*				C	O		
-16	Fuel pump special running													O		
910	Turbocharger System															
-1	Air cooler		C											O	V	Check Δp and Δt Check at every port inspection
-2	Non-return valve				C									O		
-3	Auxiliary blower													O		
-4	Butterfly valves (Not applied on Belt driven aux. blower)				C									O		
-5	Turbocharger turbine		C*						C					C,O	V,M	
	Turbocharger air filter		C											O		

V : See Volume I "Operation"

A : Adjustment to be carried out

C : Check the condition

M : See maker's instructions

O : Overhaul to be carried out

H : Check new/overhauled parts at 500, 1000 and 1500 hours

R : Parts to be replaced

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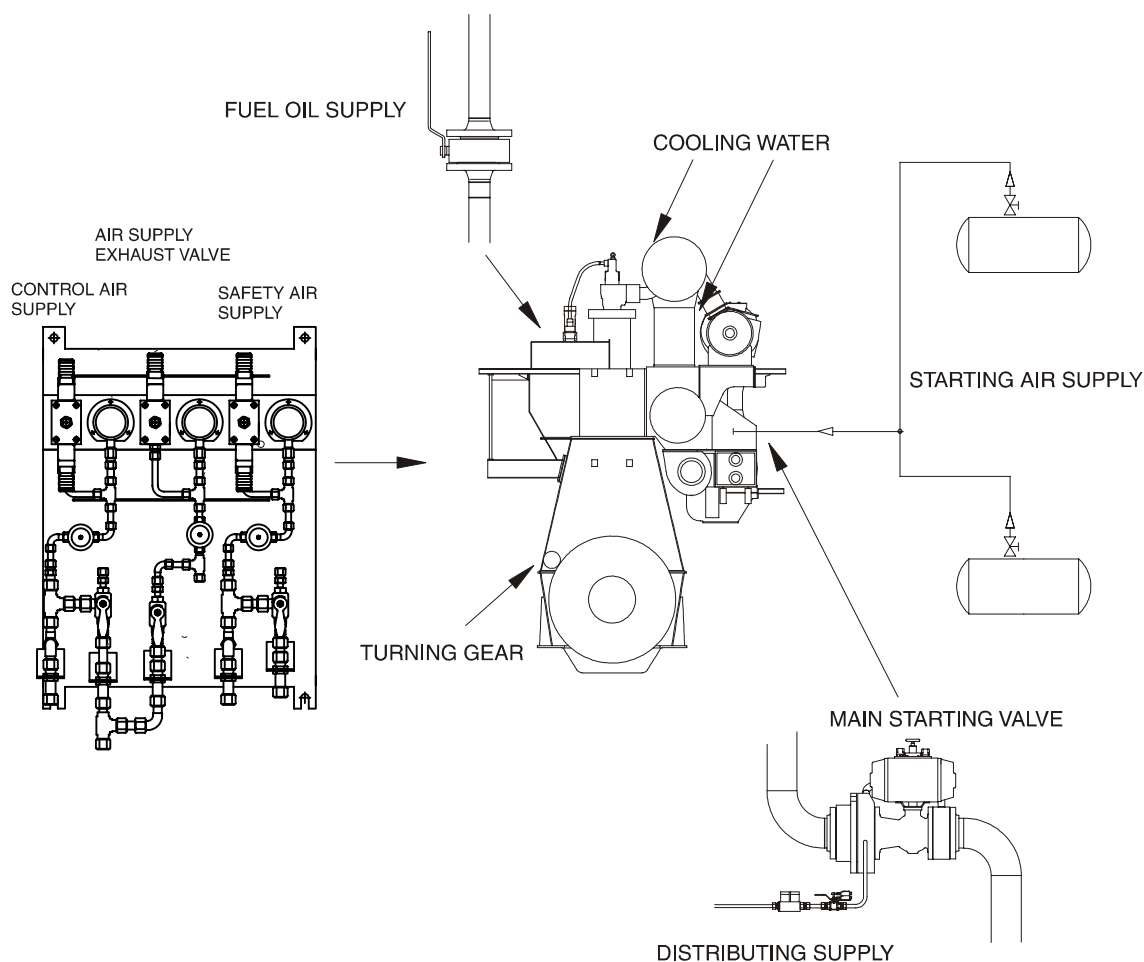
No.	Procedure	H	Regular Checks			Service interval (x 1000 hours of operation)								B	P	Remarks
			Daily	Weekly	Monthly	2	4	6	8	12	16	24	32			
	Protective grid before turbocharger								C					R	V	* Only if separate oil system * Check Δp
	Turbocharger lubricating oil water content			C*											M	
-6	Water mist catcher		C*											C,O		
911	Safety Equipment															
-1	Overpressure indicator valve								C					O		
-2	Relief valve								C					O		
-3	Scavenge air receiver safety valve								C					O		
	Functional test of alarm system for thrust bearing and slow down /shut down system	C				C									M	
	Checking and adjustment of pressure gauge								C					A	M	
	Checking and adjustment of thermometers								C					A	M	
	Checking and adjustment of thermostats								C					A	M	
	Checking and adjustment of pressurestats								C					A	M	
	Checking and adjustment of turning gear switch								C					A	M	
912	Assembly of Large Parts															
-2	Holding-down and end chock bolts	C							C							
-3	Stay bolts	C											C			
	Top bracings	C							C							
	Diaphragm in crankcase oil outlet												C	R		
	System lubricating oil water content			C											M	
	System lubricating oil bottom tank												O			Empty and clean tank
	Cooling water quality			C										A	V,M	
	Crankcase								C*							* Check for loose nuts and bearing material fragments
913	General Tools															
-1	Hydraulic tools								C					O	M	

Before maintenance work is carried out, the engine must be stopped and blocked according to the safety precautions given on the specific Data Sheet.

SAFETY PRECAUTIONS

X	Stopped engine
X	Shut off starting air supply – <i>At starting air receiver</i>
X	Block the main starting valve
X	Shut off starting air distributor/distributing system supply
X	Shut off safety air supply – <i>Not ME-engines</i>
X	Shut off control air supply
X	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
X	Engage turning gear
X	Shut off cooling water
X	Shut off fuel oil
X	Stop lubricating oil supply
X	Lock the turbocharger rotors

The sketch gives the approximate location of the valves concerned:



Safety Equipment - Recommendations

In order to perform maintenance in a safe and correct way, MAN B&W Diesel A/S recommends use of the following personal safety gear:

1.
 - Safety shoes.
 - Hearing protection.
 - Work gloves.
 - Protective head wear, such as a baseball cap with a hard protective liner.
 - Safety goggles.
 - Fall arrestor equipment.

Safety shoes, hearing protection, work gloves and protective head wear should be used when there is a risk of falling objects, loud noises and sharp edges or oily surfaces.

Use safety goggles when working with compressed air, grinders, hydraulics and when there is a risk of getting foreign objects in the eyes.

2. We strongly recommend the use of fall protection harness and arrestor equipment when working in places on the engine where there is a risk of falling or slipping.
3. This is especially needed when mounting eyebolts and hanging tackles inside the crankcase, and when working on the hydraulic pump station on some ME engines.

1.



A90060-0002C01

2.



A90060-0002C02

3.



A90060-0002C03